# CEGE: Conversational Emotion Graph Evolution*

**Bhavya V**
2023114013

**Farzana S**
2024121006

**Rudra Choudhary**
2024121012

{bhavyaram.v, farzana.s, rudra.choudhary}@research.iiit.ac.in

## Abstract

Emotion Recognition in Conversation (ERC) is essential for building empathetic dialogue systems. While prior work relies on sequential encoders or static graphs, these approaches overlook how emotional influences evolve during interaction. We introduce Conversational Emotion Graph Evolution (CEGE), a dynamic temporal graph neural network that models dialogues as evolving graphs, updating utterance representations and influence edges over time. Evaluated on the IEMOCAP dataset, CEGE — despite its dynamic design — did not outperform static baselines. This outcome suggests that dynamic modeling alone may be insufficient, highlighting the need for more effective mechanisms to capture temporal emotional evolution.

## 1 Introduction

Emotion Recognition in Conversation (ERC) is a crucial component for building emotionally intelligent and empathetic dialogue systems. While early models treated conversations as linear sequences, human dialogues exhibit complex, non-linear emotional dependencies:

- Emotional states can persist across multiple turns and speakers.
- Certain pivotal utterances can disproportionately influence subsequent emotions.
- The strength of this influence can change dynamically as the conversation progresses.

Many existing graph-based ERC models utilize static graphs, which fail to update relational dynamics as the conversation evolves. To address this limitation, we propose **Conversational Emotion Graph Evolution (CEGE)**, a dynamic graph-based model for ERC. CEGE is designed to continuously update utterance representations and their corresponding influence edges over time, aiming to better capture the temporal evolution of emotional influence among participants.

## 2 Literature Review

The field of Emotion Recognition in Conversation (ERC) has progressed from sequential models to more complex graph-based and dynamic frameworks. An early influential model, DialogueRNN (Majumder et al., 2019), utilized recurrent neural networks with attention mechanisms to capture speaker-specific dependencies. Subsequently, DialogueGCN (Ghosal et al., 2019) advanced this paradigm by employing graph convolutional networks to model speaker relationships within a static graph structure.

While effective, such static representations cannot fully capture the evolving influence of utterances over time. This limitation prompted the development of dynamic methods. For instance, DGODE (Shou et al., 2024) leveraged graph ordinary differential equations to model continuous emotional dynamics, while ERC-DP (Wang et al., 2024) focused on modeling dynamic speaker personalities.

These advancements signify a clear trend from static to adaptive graph modeling, underscoring the importance of capturing evolving emotional dependencies in dialogue. Our work, CEGE, contributes to this line of research by exploring an explicit dynamic graph evolution mechanism.

## 3 Objective

Existing ERC approaches face several notable challenges:

- **Static context modeling:** Relationships between utterances are typically fixed after initial encoding, preventing adaptation.

*https://github.com/farzanashaju/cege

- **Limited influence tracking:** The changing significance of earlier utterances on later parts of the conversation is often not captured.

This work aims to develop a dynamic graph evolution framework for ERC that models the changing emotional influence between utterances and continuously updates the conversational context over time.

# 4 Dataset

We evaluate our proposed approach on the widely-used IEMOCAP dataset (Busso et al., 2008). This dataset contains videos of dyadic conversations featuring ten unique speakers. For standardized evaluation, only the first eight speakers from sessions one to four are used for training. Each video comprises a single dialogue segmented into utterances, which are annotated with one of six emotion labels: happy, sad, neutral, angry, excited, and frustrated. For this project, we utilize only the textual modality from this dataset.

# 5 Baselines

We select DialogueRNN (Majumder et al., 2019) and DialogueGCN (Ghosal et al., 2019) as our primary baseline models. We re-implemented these models to ensure compatibility with current libraries and to verify the performance metrics reported in the original publications. Table 1 presents the results of our baseline evaluations on the IEMOCAP dataset.

# 6 Methodology

We present **CEGE: Conversational Emotion Graph Evolution**, a novel architecture that models conversations as dynamically evolving temporal graphs. Our approach comprises three integrated components: utterance encoding, dynamic temporal graph construction, and emotion-aware classification.

## 6.1 Problem Definition

Given a conversation among $M$ speakers $\{p_1, p_2, \ldots, p_M\}$, the task is to predict the emotion label for each constituent utterance $\{u_1, u_2, \ldots, u_N\}$. Each utterance $u_i$ is spoken by speaker $p_{s(i)}$, where $s$ is a mapping from an utterance to its speaker's index. We represent each utterance $u_i$ as a feature vector $u_i \in R^{D_m}$, derived from the feature extraction process described below.

## 6.2 Utterance Encoding

We employ a Convolutional Neural Network (CNN) for textual feature extraction. To capture n-gram features from each utterance, we use three distinct convolution filters of sizes 3, 4, and 5, each with 50 feature maps. The outputs are passed through a max-pooling layer followed by a Rectified Linear Unit (ReLU) activation. These activations are then concatenated and fed into a 100-dimensional dense layer to produce the final textual utterance representation. This network is pre-trained at the utterance level using emotion labels. The training loss curve is shown in Figure 2.

## 6.3 Sequential Context Encoding

Since conversations are inherently sequential, contextual information flows chronologically. To capture this flow, we process the sequence of utterance embeddings with a single-layer Long Short-Term Memory (LSTM) network. This approach differs from some prior works that use bidirectional GRUs but allows for a more natural causal encoding of context. This initial encoding is speaker-agnostic, as the utterances are encoded without speaker-specific parameters at this stage.

## 6.4 Graph Representation and Temporal Memory

A conversation with $N$ utterances is represented as a directed graph

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{R}, \mathcal{W})$$

where:

- $\mathcal{V} = \{v_1, v_2, \ldots, v_N\}$ is the set of nodes, where node $v_i$ corresponds to utterance $u_i$.
- $\mathcal{E}$ is the set of directed edges, where an edge connects a source node $v_j$ to a target node $v_i$.
- $\mathcal{R}$ is the set of relation types, categorizing edges by speaker dependencies or temporal order.
- $\mathcal{W} = \{\alpha_{ij}\}$ is the set of edge weights, where $\alpha_{ij} \in [0, 1]$ quantifies the influence of utterance $j$ on utterance $i$.

**Temporal Memory Modules** Each node $v_i$ maintains a temporal memory state $\mathbf{m}_i^t$ at time step $t$, which summarizes the evolving emotional and contextual information for that utterance:

$$\mathbf{m}_i^t = \text{LSTM}\left(\mathbf{m}_i^{t-1}, \mathbf{g}_i, \text{context}^t\right)$$

Table 1: Performance comparison of DialogueRNN and DialogueGCN on the IEMOCAP dataset. Metrics shown are per-class and weighted average accuracy and F1-score.

| Methods | Happy | | Sad | | Neutral | | Angry | | Excited | | Frustrated | | Average(w) | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Acc. | F1 | Acc. | F1 | Acc. | F1 | Acc. | F1 | Acc. | F1 | Acc. | F1 | Acc. | F1 |
| DialogueRNN | 24.23 | 32.84 | 74.82 | 78.80 | 56.77 | 58.11 | 63.15 | 64.32 | 80.13 | 70.97 | 60.89 | 57.99 | 63.07 | 61.83 |
| DialogueGCN | 40.34 | 42.37 | 88.73 | 84.00 | 61.23 | 63.05 | 66.89 | 63.76 | 65.01 | 63.01 | 63.77 | 67.01 | 64.85 | 63.66 |

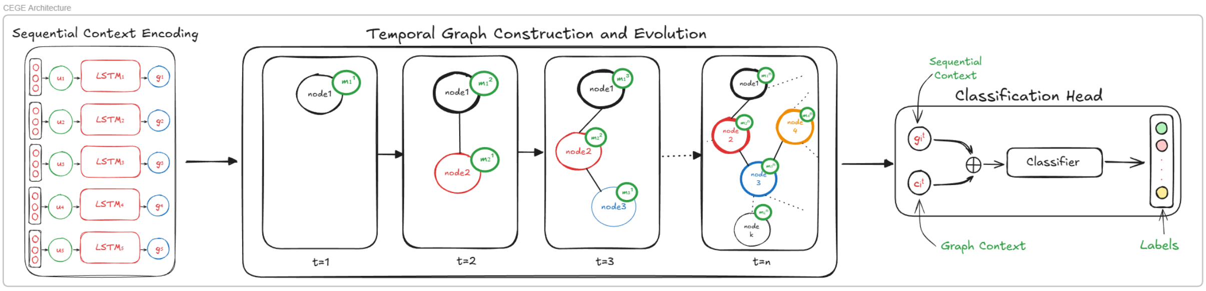


Figure 1: High Level CEGE Architecture

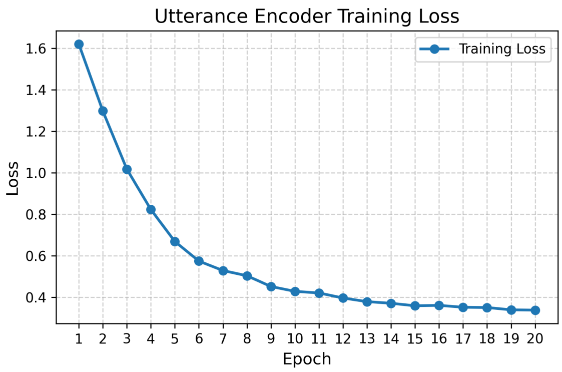


Figure 2: Utterance Encoding CNN Loss Curve.

where $\mathbf{g}_i$ is the context-aware embedding of utterance $i$, and context$^t$ includes the current speaker state $\mathbf{Q}^t_{s(i)}$ and the global conversation state $\mathbf{C}^t$.

**Graph Evolution** At each new utterance at time $t$, the graph is updated via three steps:

1. **Temporal State Change:** We compute a vector, $\Delta_t$, representing the overall change in the conversation's state. This is achieved by feeding the mean of previous memory states and the current speaker's state into a multilayer perceptron (MLP).

$$\Delta_t = \text{MLP}_{\text{analyze}}\left(\text{mean}(\mathbf{M}^{t-1}), \mathbf{Q}^t_s\right)$$

2. **Edge Weight Update:** The weight of the edge between nodes $i$ and $j$ is updated. This update considers the previous weight, the global conversational change $\Delta_t$, and the current states of the two speakers involved. A learned function determines the magnitude of the update.

3. **Edge Pruning:** To reduce noise from irrelevant historical interactions, connections with weights below a predefined threshold are pruned by setting their weight to a near-zero value.

**Note:** The temporal state change, $\Delta_t$, is a single global vector used to update all edges in the graph, providing a consistent signal of conversational shift.

### 6.5 Temporal Graph Convolution and Attention

At each time step $t$, the node features are passed through two temporal Graph Convolutional Network (GCN) layers.

**Layer 1:**

$$\mathbf{h}_i^{(1),t} = \sigma\Bigg(\sum_{r \in R^t}\sum_{j \in \mathcal{N}_i^r} \frac{\alpha_{ij}^t}{c_{i,r}} W_r^{(1)}[\mathbf{g}_j||\mathbf{m}_j^t] + W_0^{(1)}[\mathbf{g}_i||\mathbf{m}_i^t]\Bigg)$$

where $\mathbf{h}_i^{(1),t}$ is the first-layer hidden feature, $\alpha_{ij}^t$ is the dynamic edge weight, $[\mathbf{g}_j||\mathbf{m}_j^t]$ is the concatenation of the utterance embedding and its temporal memory, and $W_r^{(1)}$ and $W_0^{(1)}$ are learnable weight matrices.

**Layer 2:** The second layer aggregates information from the updated neighborhood:

$$\mathbf{h}_i^{(2),t} = \sigma\left(\sum_{j\in\mathcal{N}_i} W^{(2)}\mathbf{h}_j^{(1),t} + W_0^{(2)}\mathbf{h}_i^{(1),t}\right)$$

**Temporal Attention:** To focus on the most relevant historical context, attention weights are computed over previous memory states:

$$\beta_i^t = \mathrm{softmax}\left((\mathbf{h}_i^{(2),t})^T W_\beta[\mathbf{m}_1^t, ..., \mathbf{m}_{i-1}^t]\right)$$

A temporal decay factor is applied to prioritize more recent utterances:

$$\beta_{ik}^t = \beta_{ik}^t \cdot \exp(-\lambda_{\text{decay}} \cdot (t-k))$$

where $\lambda_{\text{decay}}$ is a hyperparameter controlling the decay rate. The final context vector $\mathbf{c}_i^t$ is a weighted sum of past memory states:

$$\mathbf{c}_i^t = \sum_{k=1}^{i-1} \beta_{ik}^t \cdot \mathbf{m}_k^t$$

### 6.6 Training and Inference

The model is trained by minimizing a composite loss function:

$$\begin{aligned}\mathcal{L} = \mathcal{L}_{\text{CE}} &+ \lambda_{\text{reg}}\|\theta\|_2^2 \\ &+ \lambda_{\text{temp}}\sum_i \|\mathbf{m}_i^t - \text{stop\_grad}(\mathbf{m}_i^{t-1})\|_2^2\end{aligned}$$

where:

- $\mathcal{L}_{\text{CE}}$ is the standard **cross-entropy loss** for emotion classification.
- $\lambda_{\text{reg}}\|\theta\|_2^2$ is an **L2 regularization term** to prevent overfitting by penalizing large parameter values ($\theta$).
- $\lambda_{\text{temp}}\sum_i \|\cdot\|_2^2$ is a **temporal consistency loss** that encourages smooth evolution of memory states by penalizing drastic changes between consecutive time steps. The stop_grad operation treats the previous state as a fixed target.

**Inference:** During testing, the model processes conversations sequentially, updating the graph and memory states for each utterance before making a prediction.

## 7 Results

Table 2 shows the accuracy and F1 scores for our proposed CEGE model. Compared to the weighted average accuracy of the DialogueGCN baseline (64.85%), our model achieves an accuracy of 63.52%. This result indicates that, for the IEMOCAP dataset, the introduction of dynamic graph components and temporal memory modules did not lead to a performance improvement over a strong static graph baseline. The per-class results show varied performance, with high accuracy on ‘Happy‘ and ‘Excited‘ emotions but significantly lower performance on ‘Neutral‘.

Table 2: Performance Metrics for the full CEGE model.

| **Test Metrics** | |
|---|---|
| **Metric** | **Score (%)** |
| Overall Accuracy | 63.52 |
| Weighted F1 | 62.38 |
| Macro F1 | 61.12 |
| **Per-Class Performance** | |
| **Class** | **Accuracy / F1 (%)** |
| Neutral | 28.47 / 38.32 |
| Happy | 84.08 / 81.75 |
| Sad | 54.43 / 57.42 |
| Angry | 62.94 / 58.15 |
| Excited | 88.63 / 75.50 |
| Frustrated | 53.28 / 55.62 |

### 7.1 Ablation Studies

To better understand the contribution of each component in our model, we conducted a series of ablation studies. We systematically removed the temporal memory module, the dynamic edge updates, and the temporal consistency loss. The results are presented in Tables 3, 4, and 5.

## 8 Analysis

The results reveal a clear but counter-intuitive outcome: every dynamic component added to the graph-based framework degraded performance instead of improving it. The full CEGE model achieved 63.52% accuracy — lower than all its ablations and the DialogueGCN baseline (64.85%).

Figure 3 helps explain this drop. The model shows strong **temporal inertia**: it tends to persist with one emotion even when the ground truth shifts. This suggests the temporal mechanisms, meant to capture context, instead make the model stick to incorrect states.

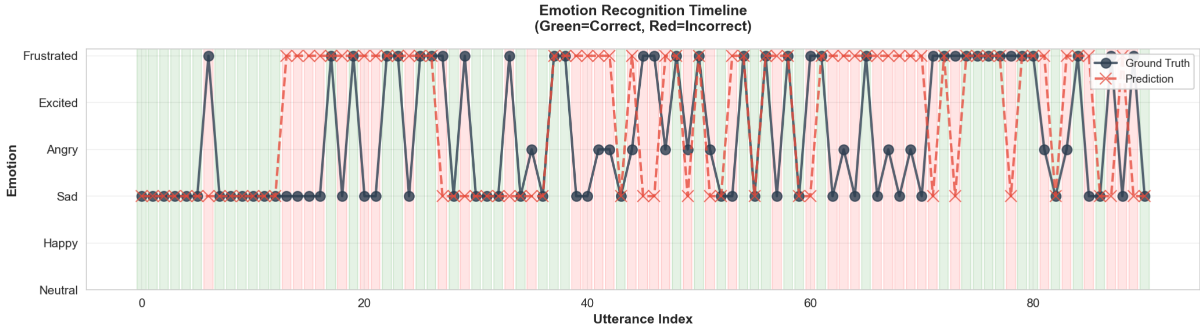


Figure 3: Emotion recognition timeline for a sample conversation. The solid dark line represents the ground truth emotion, while the dashed red line shows the model's prediction. This visualization highlights the model's tendency to maintain a predicted emotion even when the ground truth fluctuates.

Table 3: Ablation Study: Temporal Memory Module.

| **Detailed Test Metrics** | |
|---|---|
| **Metric** | **Score (%)** |
| Overall Accuracy | 64.51 |
| Weighted F1 | 64.80 |
| Macro F1 | 64.97 |
| **Per-Class Performance** | |
| **Class** | **Accuracy / F1 (%)** |
| Neutral | 70.14 / 58.21 |
| Happy | 77.14 / 80.43 |
| Sad | 52.34 / 56.46 |
| Angry | 63.53 / 60.50 |
| Excited | 67.22 / 73.36 |
| Frustrated | 64.83 / 60.84 |

Table 4: Ablation Study: Dynamic Edge Updates

| **Detailed Test Metrics** | |
|---|---|
| **Metric** | **Score (%)** |
| Overall Accuracy | 63.83 |
| Weighted F1 | 63.88 |
| Macro F1 | 64.43 |
| **Per-Class Performance** | |
| **Class** | **Accuracy / F1 (%)** |
| Neutral | 68.75 / 60.37 |
| Happy | 78.37 / 79.34 |
| Sad | 48.18 / 54.17 |
| Angry | 65.29 / 60.49 |
| Excited | 66.22 / 72.00 |
| Frustrated | 65.88 / 60.19 |

Figure 4 illustrates another issue: **contextual drift**. Tracking a single utterance over time shows that as the evolving graph updates with new inputs, the correct label's probability decays while an incorrect one rises — evidence that added context introduces noise rather than clarity.

Table 5: Ablation Study: Temporal Consistency Loss

| **Detailed Test Metrics** | |
|---|---|
| **Metric** | **Score (%)** |
| Overall Accuracy | 64.02 |
| Weighted F1 | 62.68 |
| Macro F1 | 61.20 |
| **Per-Class Performance** | |
| **Class** | **Accuracy / F1 (%)** |
| Neutral | 25.69 / 35.24 |
| Happy | 84.49 / 82.14 |
| Sad | 56.51 / 58.02 |
| Angry | 61.76 / 59.66 |
| Excited | 90.97 / 75.98 |
| Frustrated | 52.76 / 56.15 |

These patterns align with ablation results:

**Impact of the Temporal Module:** Removing the temporal memory module yielded the highest accuracy (64.51%; Table 3), indicating that the LSTM-based memory caused the temporal inertia seen in Figure 3. Instead of leveraging useful context, it overemphasized past predictions, hindering adaptation to new emotional cues.

**Impact of Dynamic Edges:** Eliminating dynamic edge updates (static graph, 63.83% accuracy; Table 4) resolved the contextual drift illustrated in Figure 4. The dynamic mechanism's updates destabilized earlier representations, injecting noise from later, often unrelated, utterances.

**Impact of Temporal Loss:** Removing the temporal consistency loss improved performance (64.02%; Table 5). Designed to smooth transitions between memory states, this constraint was overly rigid, suppressing abrupt emotional shifts and reinforcing the inertia seen in Figure 3.

Overall, each dynamic component introduced instability or rigidity, ultimately reducing performance. The results indicate that explicitly modeling temporal evolution, at least with this architecture, does not benefit emotion recognition on IEMOCAP.

## 9 Conclusion

In this paper, we introduced CEGE, a dynamic temporal graph network for Emotion Recognition in Conversation, aimed at modeling the evolving nature of emotional influence. However, experiments on the IEMOCAP dataset showed that our dynamic model did not outperform the static baseline, DialogueGCN. Ablation studies further revealed that each dynamic component incrementally reduced performance. These findings offer a valuable negative result, indicating that added dynamic complexity does not necessarily improve ERC and underscoring the strength of well-tuned static graph models.

## 10 Challenges and Limitations

As our first attempt at graph-based modeling, the main challenge lay in introducing temporality. Moving from static to evolving graph structures was conceptually difficult, and recalculating the entire graph at each time step was computationally infeasible. These constraints led us to adopt simplified incremental updates, which may have been insufficient to capture full conversational dynamics.

## 11 Future Work

The current model can be improved and expanded in several key directions:

- **Multimodality:** The current text-only model could be enhanced by incorporating parallel encoders for audio and video.
- **Alternative Temporal Mechanisms:** The performance issues with LSTMs suggest exploring more advanced temporal architectures.
- **Hyperparameter Optimization:** A thorough tuning of key hyperparameters is warranted.
- **Cross-Dataset Evaluation:** To ensure generalizability, the model should be trained and evaluated on additional ERC datasets, such as MELD, EmoryNLP, and AVEC.

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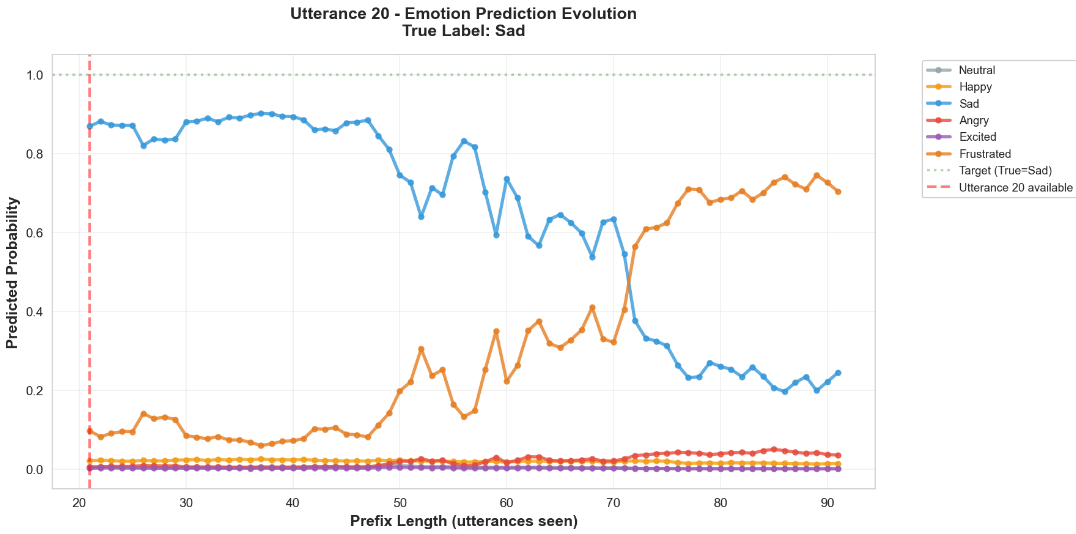


Figure 4: Evolution of predicted emotion probabilities for a single utterance as the conversation progresses. Initially, the model correctly predicts ‘Sad’ with high confidence. However, as more future utterances are observed, the dynamic graph mechanism erroneously shifts confidence toward ‘Frustrated,’ eventually leading to a misclassification. This illustrates how dynamic context updates can introduce noise and degrade earlier correct predictions.